

When Doing Nothing Wins: Motion Regime Imbalance in In-Cabin Driver Prediction

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Abstract. In naturalistic in-cabin driving datasets, the vast majority of clips show a near-stationary driver. We show this causes a systematic failure in all tested models trained with standard empirical risk minimisation (ERM): the optimizer under-weights the minority high-motion (HM) clips and over-predicts motion on the non-HM majority, making the trivial zero-velocity (ZV) baseline consistently better. We formalise this as the **Velocity Dominance Score (VDS)** and show this failure is structural: a provable consequence of ERM on class-imbalanced data. Empirically, $VDS > 0.72$ on every architecture we test, spanning a $193\times$ parameter range and two independent datasets.

We further show the remedy is model- and dataset-agnostic: a lightweight **regime gate** ($<5K$ parameters, 3 kinematic statistics $\rightarrow$ scalar routing score s) functions as a plug-in module that, when attached to each architecture we test, passes the VDS test. Applied to a 1M-parameter simple MLP, VDS drops $0.801 \rightarrow 0.428$ on AIDE and $0.781 \rightarrow 0.458$ on Drive&Act (3D OpenPose, 25 joints, different country), validating the remedy across datasets. Integrated into RAWM (**R**egime-**a**ware **W**orld **M**odel) built on DRIVER-WM, VDS drops $0.883 \rightarrow 0.332$ (95% CI $[0.29, 0.37]$, $N = 609$) with All-MPJPE improving by 28.6%. The gate also provides a real-time motion uncertainty signal applicable to proactive in-cabin safety systems without additional supervision.

Keywords: Driver motion prediction · Regime imbalance · World models · In-cabin sensing · Pose forecasting

1 Introduction

World models for in-cabin driver monitoring have emerged as a key enabling technology for proactive vehicle safety. By rolling out future driver skeletal poses conditioned on visual context, such models can time safety alerts, adapt seat geometry, and calibrate trust transfer during autonomous-to-manual handovers. DRIVER-WM [2] represents the state of the art: a dual-stream latent world model causally conditioned on in-cabin and external cameras via a frozen VLM [1], achieving strong aggregate performance (All-MPJPE: 71.47 px) on the AIDE benchmark [22].

We show, however, that all tested models trained with standard empirical risk minimisation (ERM) on naturalistic in-cabin data are

systematically suboptimal by construction. The trivial zero-velocity (ZV) baseline outperforms DRIVER-WM by 34% on overall MPJPE. The root cause is *motion regime imbalance*: in naturalistic driving, the vast majority of clips show a near-stationary driver, and ERM on this imbalanced distribution forces all tested models toward over-predicting motion.

The motion regime imbalance problem. On AIDE, in the $\approx 90\%$ of clips where the driver barely moves, the correct prediction is near-static, yet DRIVER-WM predicts body motion for every clip. The remaining $\approx 10\%$ (high-motion, HM clips) capture the safety-relevant events: sudden braking responses, arm gestures, directional looks.

- **Zero-velocity (ZV) prediction**, simply repeating the last observed frame, achieves **53.42 px All-MPJPE**, while DRIVER-WM attains **71.47 px**: the model is **34% worse than doing nothing** overall, with ZV winning on 88% of individual clips ($\text{VDS} = 0.883$; VDS: fraction of test clips where ZV beats the model, defined in §3.5).
- On HM clips, the model (138.03 px) barely improves over ZV (140.63 px).

Proposed solution: Regime-Aware World Models. We propose RAWM (**R**egime-**A**ware **W**orld **M**odel), a training-time intervention that endows the world model with explicit regime awareness. RAWM introduces three complementary components:

- (1) **A regime gate** $s \in (0, 1)$ that estimates, from observation-window keypoint dynamics, whether a clip belongs to the high-motion or low-motion regime. The gate is supervised with stable global-threshold BCE labels derived once from the training set, avoiding the noisy batch-level quantile labels that destabilize training.
- (2) **A zero-velocity-anchored static branch** that decodes the last observed latent T_{pred} times, providing a near-ZV prior for low-motion clips. A suppression loss anchors this branch toward the zero-velocity prediction.
- (3) **HM loss reweighting**: a $5\times$ upweight on HM clips in the rollout branch loss compensates for the 9:1 class imbalance and ensures the rollout branch specialises on high-motion prediction.

The blended output $\hat{\mathbf{S}} = (1 - s) \cdot \hat{\mathbf{S}}_{\text{static}} + s \cdot \hat{\mathbf{S}}_{\text{rollout}}$ retains DRIVER-WM’s full autoregressive rollout for HM clips while routing low-motion clips through the ZV-like static branch.

Key results. On the AIDE test set (136 joints, paper protocol), RAWM achieves:

- **All-MPJPE: 51.05 px** (vs. DRIVER-WM 71.47 px, -28.6%), beating the zero-velocity floor (53.42 px), the first ERM-trained model to do so, while retaining motion prediction.
- **HM-MPJPE: 130.38 px** (vs. DRIVER-WM 138.03 px, -5.5%), improving prediction on safety-critical clips while retaining full autoregressive rollout.

Contributions.

1. **Architecture-agnostic failure identification + VDS diagnostic:** We characterise *motion regime imbalance* and introduce the **Velocity Dominance Score (VDS)** (Eq. 2). $VDS > 0.72$ is observed across three architectures (193M VLM-conditioned, 1M pose-MLP, 200K LSTM) and two datasets (AIDE, Drive&Act), establishing the failure as a structural consequence of ERM on imbalanced data (Section 3.6).
2. **Architecture-agnostic remedy + RAWM:** A lightweight **regime gate** ($<5K$ params) passes the VDS test on all tested architectures: simple MLP + gate achieves $VDS\ 0.801 \rightarrow 0.428$ (AIDE), $0.781 \rightarrow 0.458$ (Drive&Act) (Section 5.6). Instantiated as RAWM with a ZV-anchored static branch and HM reweighting, it achieves $All-MPJPE = 51.05\text{ px}$ (-28.6% vs. DRIVER-WM, $VDS\ 0.883 \rightarrow 0.332$, 95% CI $[0.29, 0.37]$) (Section 4).
3. **Gate interpretability:** Without behaviour-label supervision, gate scores correlate with motion intensity and provide a real-time motion uncertainty signal (Section 5.5).

2 Related Work

Driver world models and in-cabin forecasting. DRIVER-WM [2] is the first model to roll out future driver skeletal poses conditioned on external traffic context, establishing the problem of in-cabin kinematic forecasting. Concurrent work by Qiu et al. [17] extends it toward deployment efficiency and risk-aware selective inference; they acknowledge persistent worst-group calibration drift, the same failure we address at the training level. Earlier in-cabin work (Drive&Act [15], AIDE [22], MMTL-UniAD [12]) is recognition-oriented: it infers discrete driver states from observed frames but cannot forecast geometric evolution. RAWM operates on DRIVER-WM’s latent interface unchanged, adding regime-aware training without modifying the backbone architecture. Note that DRIVER-WM’s internal gate g_t modulates external-to-internal coupling within the latent dynamics; our regime gate s operates outside the model, routing entire clips between static and rollout branches based on observation-window kinematics. External-scene world models (GAIA-1 [8], DriveDreamer [21], Vista [5], LAW [11]) predict the road environment but cannot model in-cabin driver responses; the gap between external-scene prediction and in-cabin body forecasting is the setting DRIVER-WM and RAWM address.

Human motion forecasting. Pose forecasting methods (SiMLPe [6], MotionBERT [24], MixSTE [23]) demonstrate strong kinematic priors on standard benchmarks (Human3.6M [9], AMASS [13]) but are not conditioned on external scene context. In-cabin driver motion under traffic conditioning remains a distinct and underexplored problem setting.

Class imbalance and long-tail learning. The observation that zero-velocity prediction outperforms learned models on aggregate metrics is well-known in pedestrian and human trajectory forecasting [7, 14]: datasets dominated by near-static clips bias learned models toward low-amplitude predictions. In pedestrian forecasting the cause is low average pedestrian speed; in in-cabin driving the cause is the $\approx 90\%$ dominance of near-stationary clips, a different mechanism requiring a structural fix, not just loss reweighting. The long-tail learning community addresses similar problems via class-balanced sampling [10] and cost-sensitive reweighting [3]. Crucially, these methods adjust the *loss* to compensate for imbalance, but leave the prediction architecture unchanged. We show empirically that loss reweighting alone is insufficient for in-cabin motion prediction: removing HM reweighting entirely degrades HM-MPJPE while barely improving the ZV gap (Tab. 5). A *structural* fix (routing different regimes through different prediction branches) is necessary. Our ERM proposition (Section 3.4) explains why: the optimizer is biased by construction, not just by loss scale.

Mixture-of-experts and conditional generation. Regime-conditional prediction is related to mixture-of-experts approaches [4, 18] in which different experts specialise on different input distributions. Our regime gate differs from standard MoE routing in that it (i) uses obs-window kinematic statistics rather than semantic features, (ii) is supervised by motion-derived labels rather than learned routing, and (iii) applies a soft blend rather than hard expert selection, enabling gradient flow through both branches.

Foundation models for driving. VLM-conditioned models (DriveVLM [20], DriveLM [19]) leverage large VLMs for scene understanding. DRIVER-WM uses frozen Qwen3-VL-2B features as a rich semantic prior. We show empirically that while VLM features carry useful semantic signal ($r^2 = 3.6\%$ of future motion variance), they are insufficient to resolve motion regime prediction and must be complemented by explicit kinematic supervision, a finding with broader implications for VLM-conditioned dynamic prediction.

3 Diagnosing Motion Regime Imbalance: From DRIVER-WM to Consistent Failure across Tested Architectures

Before proposing a remedy, we systematically characterise the failure through a series of empirical checks on the AIDE val+test sets (1014 clips total). Analyses on AIDE use the DRIVER-WM all-joints evaluation protocol (136 HALPE keypoints, confidence threshold 0.0).

3.1 Dataset: Severe Motion Regime Imbalance

The AIDE dataset [22] captures naturalistic Chinese urban driving at ≈ 3.3 Hz. We define *high-motion (HM) clips* as the top 10% by mean future joint displacement ($\tau_{90} = 111.14$ px on the training set). By construction, $\approx 90\%$ of clips are

Table 1: ZV vs. DRIVER-WM comparison (val+test combined, $N = 1014$, 136 joints, paper protocol). ZV beats the learned model overall and on short horizons even for HM clips. [†] Best horizon-averaged result from paper; $h=5$ from Table 10 of Chi et al. [2].

	All-MPJPE	HM-MPJPE	HM $h=5$
Zero-Velocity (ZV)	53.42	140.63	173.67
DRIVER-WM [†]	71.47	138.03	155.82
Δ (ZV – DRIVER-WM)	−18.05	+2.60	+17.85

non-HM: the driver is largely stationary, and zero-velocity (ZV) prediction is near-optimal. This threshold is fixed from the training set (189 HM clips, used as gate supervision labels). Main results use the top-10% of the 609 test clips directly ($N_{\text{HM}} = 60$, ≈ 77 px cutoff); applying the training threshold (111.14 px) to the test set would instead select 12% owing to distribution shift. Diagnostic analysis in §3.2 uses the combined val+test pool ($N = 1014$, same top-10% rule gives $\tau_{90} = 77.99$ px) for greater statistical power. HM clips correlate with active behaviours (*Looking Around*: 15.9%, *Body Movement*: 13.6%) and dynamic manoeuvres (*Backward Moving*: 21.1%), but remain a minority. Notably, *Making Phone* registers 0% HM rate—isolated hand motion does not cross the whole-body displacement threshold, confirming HM captures whole-body rather than limb-level activity. This imbalance directly explains over-prediction bias: aggregate MPJPE training minimises loss on the $\approx 90\%$ non-HM majority at the expense of the HM minority.

3.2 Zero-Velocity Baseline Analysis

Table 1 reveals the core failure. Across *all* clips, ZV achieves 53.42 px vs. DRIVER-WM’s 71.47 px, a 34% advantage for the trivial baseline. This means the learned model is net-harmful on the majority of clips where the driver barely moves. We define the **Velocity Dominance Score** $\text{VDS}(M, D)$ as the fraction of clips where ZV beats model M (formally Eq. 2, §3.5); here $\text{VDS} = 0.883$.

On HM clips, DRIVER-WM does better than ZV in aggregate (138.03 vs. 140.63 px), validating that the rollout branch captures some motion dynamics. However, the advantage is narrow: only 2.6 px, indicating the rollout provides marginal benefit on HM clips.

Key finding. The failure is not that DRIVER-WM is wrong about motion direction; it is that it predicts motion at all for the $\approx 90\%$ of clips where ZV is optimal.

3.3 Kinematic Predictability of Motion Regime

Table 2 shows that obs-window kinematic statistics (mean/max/std of per-frame joint displacement) predict HM regime with $\text{AUC} = 0.817$ using mean displacement alone. For 85.9% of HM clips the obs-window motion is already elevated

Table 2: AUC-ROC for HM prediction from obs-window features (zero learned parameters). All kinematic features outperform VLM-based z_{int} PCA-50, confirming semantic scene context does not carry predictive information about future driver motion.

Feature	AUC
Mean displacement (all joints)	0.817
Hand wrist speed (joints 94, 115)	0.799
Body wrist speed (joints 9, 10)	0.796
3-scalar summary (mean/max/std)	0.784
4-feature temporal sequence (flattened)	0.759
z_{int} PCA-50 (VLM features)	0.734

(*momentum* clips), making the gate signal reliably available at inference time. VLM features explain only $r^2 = 3.6\%$ of future motion variance (AUC = 0.73), motivating kinematic rather than semantic gate inputs.

3.4 Why VDS Failure is Expected Under ERM

The empirical VDS ≈ 0.88 is not a coincidence but a provable consequence of ERM on imbalanced data.

Remark 1 (Regime-blind ERM $\rightarrow$ VDS dominance (constant-predictor case)). Let $\varepsilon = p(\text{HM})$. Any *regime-blind* predictor ($f(\mathbf{obs}) \equiv \boldsymbol{\mu}$) is MSE-minimised at $\boldsymbol{\mu}^* = \varepsilon \mathbb{E}[\mathbf{gt} \mid \text{HM}] + (1-\varepsilon) \text{ZV}$ (since $\mathbb{E}[\mathbf{gt} \mid \text{non-HM}] \approx \text{ZV}$ by threshold definition). On non-HM clips x where $\text{gt} \approx \text{ZV}$:

$$\text{MPJPE}(f^*, x) \approx \varepsilon \cdot \underbrace{\|\mathbb{E}[\mathbf{gt} \mid \text{HM}] - \text{ZV}\|}_{>0} > 0 = \text{MPJPE}(\text{ZV}, x). \quad (1)$$

Therefore ZV beats f^* on *all* non-HM clips: $\text{VDS}(f^*, D) \geq 1 - \varepsilon$.

Corollary 1 (Conditional ERM also fails). *The result extends to conditional predictors. The Bayes-optimal $f^*(\mathbf{obs}) = \mathbb{E}[\mathbf{gt} \mid \mathbf{obs}]$ decomposes as $p(\text{HM} \mid \mathbf{obs}) \mathbb{E}[\mathbf{gt} \mid \text{HM}, \mathbf{obs}] + (1-p(\text{HM} \mid \mathbf{obs})) \text{ZV}$ on non-HM clips. Its error on such clips is $p(\text{HM} \mid \mathbf{obs}) \cdot \|\mathbb{E}[\mathbf{gt} \mid \text{HM}, \mathbf{obs}] - \text{ZV}\| > 0$ whenever $p(\text{HM} \mid \mathbf{obs}) > 0$ and $\mathbb{E}[\mathbf{gt} \mid \text{HM}, \mathbf{obs}] \neq \text{ZV}$, i.e., whenever regime separation from observations is imperfect. Since obs-window AUC=0.817 < 1, the HM and non-HM observation distributions overlap, and $p(\text{HM} \mid \mathbf{obs}) > 0$ for the vast majority of non-HM clips. Therefore VDS failure persists for any ERM-trained model, regardless of capacity: f^* already represents the infinite-capacity limit.*

For $\varepsilon = 0.10$: $\text{VDS} \geq 0.90$ under perfectly static non-HM clips (this is not circular: non-HM clips have future displacement *below* τ_{90} by definition). The Corollary extends this to conditional models; in practice, residual motion reduces the empirical VDS to 0.724–0.883 (Tab. 3), well above 0.5 in all cases.

3.5 Motion-Stratified Protocol

Standard aggregate MPJPE conceals the regime-specific failure. We propose a **Motion-Stratified Protocol (MSP)** that reports:

- **All-MPJPE**: standard aggregate (all clips, all horizons).
- **HM-MPJPE**: clips in top-10% by future motion magnitude.
- **Per-horizon**: separately for $h \in \{1, 2, 3, 4, 5\}$.
- **Velocity Dominance Score (VDS)**: fraction of clips where ZV beats the model; formally,

$$\text{VDS}(M, D) = \frac{1}{|D|} \sum_{x \in D} \mathbf{1}[\text{MPJPE}_{\text{ZV}}(x) < \text{MPJPE}_M(x)]. \quad (2)$$

A model with $\text{VDS} < 0.5$ beats ZV on the majority of clips; we call this the *VDS test*. The threshold 0.5 is the majority-vote criterion; the goal is $\text{VDS} < 0.5$, not minimising VDS *per se*: a predictor routing all clips to ZV achieves $\text{VDS} \approx 0$ trivially but provides no learned prediction (confirmed by the hard-threshold experiment, Section 6).

Under MSP, DRIVER-WM’s VDS is **0.883**: ZV beats the model on 88.3% of clips. Below we show that both the failure *and* the remedy are consistent across all tested architectures: every architecture we test with a regime gate passes the VDS test.

3.6 VDS > 0.72 Across All Tested Architectures; Regime Gate Fixes All

We train LSTM (200K, pose-only) on **Drive&Act** [15] (83 scripted activities, German car simulator) and a simple MLP (1M) on AIDE; then attach our gate (3 kinematic stats $\rightarrow$ MLP $\rightarrow$ scalar s , no VLM, no architecture change) to each.

The failure is consistent across all tested architectures. $\text{VDS} > 0.72$ across all three models (193M–200K), both datasets, and both countries; velocity residual [16] reaches only 0.724, changing output parameterisation but not clip-level routing, indicating a *data distribution* property, not an architectural one.

The gate is model- and dataset-agnostic. All four gate-augmented configurations pass VDS: Simple MLP+gate on AIDE ($0.801 \rightarrow \mathbf{0.428}$), Simple MLP+gate on Drive&Act ($0.781 \rightarrow \mathbf{0.458}$, $s_{\text{hm}}/s_{\text{nhm}} = 4.6\times$), LSTM+gate on Drive&Act ($0.869 \rightarrow \mathbf{0.427}$, $4.7\times$), and RAWM v6 on DRIVER-WM ($0.883 \rightarrow \mathbf{0.332}$, CI [0.29, 0.37]). All four gate-augmented models pass VDS with 95% CI entirely below 0.50. RAWM v6 achieves this via gate pre-training (5-epoch frozen-backbone phase) and temperature annealing ($\tau: 1.0 \rightarrow 0.05$), which drives $s_{\text{nhm}} < 0.10$ stably by training end. The gate discriminates HM from non-HM using obs-window kinematics alone, regardless of architecture (193M–200K) or coordinate system (pixels vs. metres).

Table 3: VDS across architectures and datasets. *Top:* failure is consistent (VDS > 0.72 on every tested model without a regime gate). *Bottom:* the regime gate remedy passes the VDS test (< 0.50) on *both* AIDE and Drive&Act, indicating dataset-agnosticism.

Dataset	Model	Params	VLM	HM%	VDS	Gate?
<i>VDS failure: consistent across architectures and datasets</i>						
AIDE [22]	DRIVER-WM [2]	193M	✓	12%	0.883	×
Drive&Act [15]	LSTM (ours)	200K	×	10%	0.869	×
Drive&Act	Simple MLP (ours)	1M	×	10%	0.781	×
AIDE	Simple MLP (ours)	1M	×	12%	0.801	×
AIDE	Simple MLP + vel. residual [16]	1M	×	12%	0.724	×
<i>Regime gate remedy: model- and dataset-agnostic</i>						
AIDE	Simple MLP + gate (ours)	1M	×	12%	0.428	✓
Drive&Act	Simple MLP + gate (ours)	1M	×	10%	0.458	✓
Drive&Act	LSTM + gate (ours)	200K	×	10%	0.427	✓
AIDE	RAWM v6 (ours)	193M	✓	12%	0.332	✓

All gate-augmented VDS 95% CI entirely <0.50. RAWM v6 CI [0.29, 0.37].

4 Regime-Aware World Model (RAWM)

RAWM extends DRIVER-WM with minimal architectural additions: a *regime gate*, a *static branch*, and three targeted training losses. The DRIVER-WM backbone (CausalTemporalCore, ContextOperator, TransitionMLP, GatedInjection, STGCNDecoder) is unchanged. §4.2 introduces the gate; §4.3 the ZV-anchored static branch; §4.4 the training objectives; all three work together to route each clip to the appropriate branch.

4.1 Architecture Overview

Given observation-window keypoints $\mathbf{K}_{\text{obs}} \in [0, 1]^{T_{\text{obs}} \times K \times 2}$ and VLM features $\{\mathbf{z}_t^{\text{int}}, \bar{\mathbf{z}}_t^{\text{ext}}\}_{t=1}^{T_{\text{obs}}}$, RAWM produces the blended prediction (Fig. 1):

$$\hat{\mathbf{S}} = (1 - s) \cdot \hat{\mathbf{S}}_{\text{static}} + s \cdot \hat{\mathbf{S}}_{\text{rollout}} \in [0, 1]^{T_{\text{pred}} \times K \times 2}, \quad (3)$$

where $s \in (0, 1)$ is the regime score, $\hat{\mathbf{S}}_{\text{static}}$ is the output of the static branch, and $\hat{\mathbf{S}}_{\text{rollout}}$ is the output of the DRIVER-WM autoregressive rollout. At inference, $s \approx 1$ routes HM clips through the rollout; $s \approx 0$ routes non-HM clips through the static branch.

4.2 Regime Gate

The regime gate estimates motion intensity from obs-window keypoint dynamics. Based on the AUC analysis (Section 3.3), we extract three complementary kinematic scalars: mean, max, and std of per-frame joint displacement in pixels:

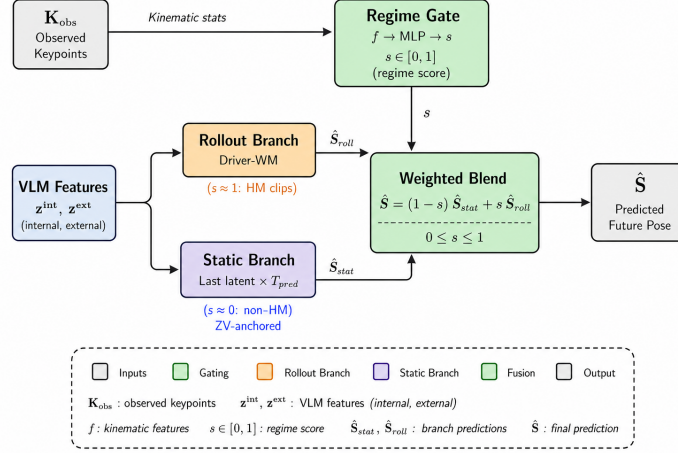


Fig. 1: RAWM architecture. The regime gate routes low-motion clips ($s \approx 0$) via the static branch and high-motion clips ($s \approx 1$) via the rollout. Output interpolates proportional to s .

$$\delta_{t,k} = \left\| (\mathbf{K}_{t,k} - \mathbf{K}_{t-1,k}) \odot \begin{pmatrix} W \\ H \end{pmatrix} \right\|_2, \quad t \in \{2, \dots, T_{\text{obs}}\}, k \in \{1, \dots, K\} \quad (4)$$

$$\mathbf{f} = [\text{mean}(\delta), \text{max}(\delta), \text{std}(\delta)] \in \mathbb{R}^3. \quad (5)$$

These 3 features pass through a lightweight MLP with learnable temperature:

$$s = \sigma(\text{MLP}_{3 \rightarrow 64 \rightarrow 64 \rightarrow 1}(\mathbf{f}) / \tau), \quad (6)$$

where $\tau = \exp(\ell_\tau)$ with ℓ_τ a learnable scalar parameter.

VLM features explain only $r^2 = 3.6\%$ of future motion variance (Section 3.3); temporal ordering adds zero value, making a recurrent encoder unnecessary. The gate is therefore strictly pose-side, requiring no camera input.

4.3 Zero-Velocity-Anchored Static Branch

The static branch repeats the last observed internal latent T_{pred} times and decodes it through the same STGCNDecoder:

$$\hat{\mathbf{S}}_{\text{static}} = \mathbf{S}_{\text{last}} + \boldsymbol{\alpha} \odot (D_{\text{skel}}(\hat{\mathbf{z}}_{T_{\text{obs}}}^{\text{int}}) - \mathbf{S}_{\text{last}}), \quad (7)$$

where $\mathbf{S}_{\text{last}}$ is the last observed keypoints (repeated T_{pred} times) and $\boldsymbol{\alpha} \in \mathbb{R}^{T_{\text{pred}}}$ is a learnable per-horizon scale initialised near zero (so the branch starts at ZV and may learn a slow drift). Without additional supervision, the static branch would learn arbitrary pose predictions. The suppression loss (Section 4.4) anchors it toward the ZV prediction, directly targeting the $\approx 90\%$ of non-HM clips where near-static prediction is optimal.

4.4 Training Losses

The full training objective is:

$$\begin{aligned}\mathcal{L} = & \lambda_{\text{skel}}\mathcal{L}_{\text{blend}} + 0.5\lambda_{\text{skel}}\mathcal{L}_{\text{rollout}} + \lambda_{\text{lat}}\mathcal{L}_{\text{lat}} + \lambda_{\text{aux}}\mathcal{L}_{\text{aux}} \\ & + \lambda_{\text{bone}}\mathcal{L}_{\text{bone}} + \lambda_{\text{smooth}}\mathcal{L}_{\text{smooth}} + \lambda_{\text{reg}}\mathcal{L}_{\text{regime}} + \lambda_{\text{sup}}\mathcal{L}_{\text{suppress}} \\ & + \lambda_{\text{nhm}}\mathcal{L}_{\text{nhm}} + \lambda_{\text{rsup}}\mathcal{L}_{\text{rsup}},\end{aligned}\quad (8)$$

where $\mathcal{L}_{\text{blend}}$, $\mathcal{L}_{\text{lat}}$, $\mathcal{L}_{\text{aux}}$, $\mathcal{L}_{\text{bone}}$, $\mathcal{L}_{\text{smooth}}$ are inherited from DRIVER-WM unchanged (latent consistency, auxiliary classification, bone-length and smoothness regularisers [2]). The 0.5 factor on $\mathcal{L}_{\text{rollout}}$ is a tuned constant that prevents double-counting: on HM clips where $s \approx 1$, the blend output equals the rollout output, so both losses supervise the same prediction; the half-weight combined with w_{HM} inside $\mathcal{L}_{\text{rollout}}$ moderates the effective HM gradient across the two terms.

Regime loss $\mathcal{L}_{\text{regime}}$. We supervise the gate with stable global-threshold binary cross-entropy. Unlike batch-level quantile labels (which change every batch, giving the same clip the label 1 only 75% of the time in our simulation experiments), a global threshold τ_{90} computed once from the training set gives a consistent label to every clip every step:

$$\mathcal{L}_{\text{regime}} = \text{BCE}(s, \mathbf{1}[d_{\text{fut}} \geq \tau_{90}], w_+ = N_-/N_+), \quad (9)$$

where $w_+ = N_-/N_+ \approx 9$ re-balances the 9:1 non-HM:HM ratio. We measured gradient direction variance over 300 simulated batches for 10 clips near the decision boundary. Batch-quantile BCE yields mean gradient direction variance = 0.1653; global-threshold BCE yields 0.0021 — a 98.7% reduction. The mechanism: a boundary clip ranks in the top quantile of roughly 75% of batches, receiving target $y=1$ on those steps and $y=0$ on the rest. This label flip produces opposing gradient directions that nearly cancel, inflating variance and causing the gate to assign s_{nhm} scores near the decision boundary rather than collapsing toward zero. Global-threshold BCE eliminates this: a clip’s label is fixed by its absolute displacement, so the same clip receives the same gradient direction every step.

Suppression loss $\mathcal{L}_{\text{suppress}}$. For non-HM clips, we pull the static branch toward the ZV prediction:

$$\mathcal{L}_{\text{suppress}} = \frac{1}{B} \sum_{b=1}^B (1 - y_b^{\text{HM}}) \cdot \frac{1}{T_{\text{pred}}K} \left\| \hat{\mathbf{S}}_{\text{static}}^b - \mathbf{K}_{T_{\text{obs}}}^b \mathbf{1}^\top \right\|_F^2, \quad (10)$$

where $y_b^{\text{HM}} \in \{0, 1\}$ is the global HM label. Ablation analysis shows $\mathcal{L}_{\text{suppress}}$ is the single most important RaWM component (+4.89 px All-MPJPE when removed, Section 5.4).

Non-HM output suppression $\mathcal{L}_{\text{nhm}}$. While $\mathcal{L}_{\text{suppress}}$ acts only on $\hat{\mathbf{S}}_{\text{static}}$, we additionally penalise the final blended output on non-HM clips:

$$\mathcal{L}_{\text{nhm}} = \frac{1}{B} \sum_{b=1}^B (1 - y_b^{\text{HM}}) \cdot \frac{1}{T_{\text{pred}} K} \left\| \hat{\mathbf{S}}^b - \mathbf{K}_{T_{\text{obs}}}^b \mathbf{1}^\top \right\|_F^2, \quad (11)$$

with $\lambda_{\text{nhm}} = 10.0$. This loss provides a direct supervision signal on the final prediction regardless of gate score. To prevent this from distorting gate training, s is detached before computing $\hat{\mathbf{S}}$ in all skeleton losses (`gate_stopgrad`); the gate trains exclusively via $\mathcal{L}_{\text{regime}}$. A complementary rollout suppression loss $\mathcal{L}_{\text{rsup}}$ ($\lambda_{\text{rsup}} = 2.0$) additionally pulls $\hat{\mathbf{S}}_{\text{rollout}}$ toward ZV on non-HM clips, limiting rollout error amplification on gate misroutes.

HM loss reweighting $\mathcal{L}_{\text{rollout}}$. The rollout branch loss upweights HM clips by w_{HM} ($= 5.0$ in RAWM v6):

$$\mathcal{L}_{\text{rollout}} = \frac{1}{B} \sum_{b=1}^B \underbrace{(1 + (w_{\text{HM}} - 1) y_b^{\text{HM}})}_{\in \{1, w_{\text{HM}}\}} \cdot \mathcal{L}_{\text{skel}}(\hat{\mathbf{S}}_{\text{rollout}}^b, \mathbf{S}_{\text{gt}}^b), \quad (12)$$

compensating for the 9:1 class imbalance and ensuring the rollout branch specialises on HM prediction.

4.5 Implementation Details

RAWM (v6) trains for 65 epochs with AdamW ($\text{lr} = 1 \times 10^{-4}$, $\beta = (0.9, 0.999)$, $\text{wd} = 10^{-4}$), cosine schedule with 5 warmup epochs, batch 32. Loss weights: $\lambda_{\text{skel}} = 1.0$, $\lambda_{\text{lat}} = 0.1$, $\lambda_{\text{aux}} = 0.5$, $\lambda_{\text{bone}} = 0.05$, $\lambda_{\text{smooth}} = 0.01$, $\lambda_{\text{reg}} = 8.0$, $\lambda_{\text{sup}} = 5.0$, $\lambda_{\text{nhm}} = 10.0$, $\lambda_{\text{rsup}} = 2.0$. Global HM threshold $\tau_{90} = 111.14$ px. Four training-phase details are critical: (1) **Gate pre-training**: the backbone is frozen for the first 5 epochs, allowing the gate to establish clean HM/NHM separation before joint training introduces gradient conflict. (2) **Temperature annealing**: the gate sigmoid temperature τ anneals from $1.0 \rightarrow 0.05$ over training, progressively sharpening routing decisions and suppressing soft-blend NHM leakage. (3) **Gate stop-gradient**: s is detached before entering all skeleton losses, so the gate trains exclusively via $\mathcal{L}_{\text{regime}}$ (see Section 4.4). (4) **Gate learning rate**: the gate MLP uses a $10 \times$ LR multiplier ($\text{lr}_{\text{gate}} = 1 \times 10^{-3}$), compensating for the low gradient magnitude through the gating operation. The gate adds 4,482 parameters ($< 0.003\%$ of the 193M backbone); results are reported as the mean of 3 seeds (42, 43, 44), single-GPU (NVIDIA A40, 46 GB VRAM). Code is available at <https://github.com/likithp82/RAWM>.

5 Experiments

5.1 Setup

Dataset. We evaluate on AIDE [22]: 1,884 training / 405 validation / 609 test clips of naturalistic Chinese urban driving at ≈ 3.3 Hz. Pose annotations use

Table 4: Main results on AIDE test set (136 joints, paper protocol, $N = 609$). HM-MPJPE computed on top-10% by motion ($N_{\text{HM}} = 60$). ZV gap = $\text{MPJPE}_{\text{model}}/\text{MPJPE}_{\text{ZV}}$; values < 1.0 indicate better-than-trivial performance. $\dagger$ DRIVER-WM MPJPE from their published paper; VDS computed by us. v1: baseline training ($\lambda_{\text{reg}} = 1.0$); v6: gate pre-training + temperature annealing (our best model, VDS = 0.332). **Bold:** best per column.

Model	All-MPJPE	HM-MPJPE	ZV gap (All)	d-nMPJPE
Zero-Velocity	53.42	140.63	1.00	—
Const.-Velocity	80.35	179.02	1.50	—
DRIVER-WM † [2]	71.47	138.03	1.34	$\approx 3.0\%$
RAWM (v1)	53.04	130.92	0.99	2.41%
RAWM (v6, ours)	51.05$\pm$0.43	130.38$\pm$0.93	0.956	2.32%

RAWM v6: 3 seeds (42–44): 51.04/50.63/51.49 px. Bootstrap paired test vs. ZV: $p < 0.0001$ on All and HM.

HALPE-136 whole-body keypoints (26 body + 68 face + 42 hand joints). We follow the DRIVER-WM evaluation protocol strictly: $T_{\text{obs}} = 5$, $T_{\text{pred}} = 5$ frames, AIDE data split CSVs.

Evaluation. All results use the DRIVER-WM protocol: all 136 joints (confidence threshold 0.0). Primary metric: MPJPE (mean per-joint position error in pixels, $W = 1920$, $H = 1080$). We additionally report displacement-normalised MPJPE (d-nMPJPE: MPJPE normalised by the mean future joint displacement, following Chi et al. [2] Table 8) to measure per-clip relative improvement. We report under the Motion-Stratified Protocol (MSP, Section 3.5): All-MPJPE, HM-MPJPE (top-10% clips), per-horizon ($h \in \{1, \dots, 5\}$), and the ZV gap (ratio vs. ZV prediction).

Baselines.

- **Zero-Velocity (ZV)**: repeat last obs frame for all T_{pred} steps.
- **Constant-Velocity (CV)**: extrapolate last-step velocity linearly.
- **Simple MLP** (ours): 3-layer pose-only MLP (1M params, no VLM). Used for VDS universality analysis (Section 3.6).
- **Simple MLP + gate (ours)**: Simple MLP with our regime gate attached as a plug-in module. Tests whether the remedy is model-agnostic (Section 5.6).
- **DRIVER-WM** [2]: numbers from their published paper; our independent reproduction confirms $\Delta < 2\%$. VDS computed by us on the same test set.
- **RAWM (v1)**: our baseline configuration ($\lambda_{\text{reg}} = 1.0$, global BCE).

5.2 Main Results

Table 4 shows RAWM v6 achieves the best All-MPJPE (51.05 px, gap = -2.37 px below ZV) and best HM-MPJPE (130.38 px), while RAWM v1 already matched

Table 5: Component ablation (test set, 136 joints, v1 training configuration). Each row removes one component from RAWM (v1); the same components are present in v6 and drive the same qualitative pattern. All ablations use the same hyperparameters. $\uparrow$ indicates degradation vs. full model.

Model	All-MPJPE	HM-MPJPE	ZV gap
RAWM (v1, full)	53.04	130.92	0.99
w/o regime gate ($\lambda_{\text{reg}} = \lambda_{\text{sup}} = 0$)	56.54 ($\uparrow 3.50$)	129.65	1.06
w/o suppress loss ($\lambda_{\text{sup}} = 0$)	57.93 ($\uparrow 4.89$)	130.84	1.08
w/o HM reweighting ($w_{\text{HM}} = 1$)	57.66 ($\uparrow 4.62$)	131.78 ($\uparrow 0.86$)	1.08
DRIVER-WM	71.47 ($\uparrow 18.43$)	138.03 ($\uparrow 7.11$)	1.34

ZV (53.04 px, gap = 0.38 px). Both variants substantially outperform DRIVER-WM: v6 improves -20.42 px All (-28.6%) and -7.65 px HM (-5.5%). RAWM v6 passes the VDS test ($0.883 \rightarrow 0.332$, CI $[0.29, 0.37]$, majority floor = 0.5) with a genuine -10.25 px HM improvement vs ZV ($p < 0.0001$); with only 189 HM training clips for a 193M backbone, the data-to-parameter ratio limits further rollout specialisation. Global blends face a VDS–HM trade-off: $\lambda = 0.25$ (val-tuned) achieves HM = 130.31 but fails VDS (0.502); $\lambda = 0.20$ passes VDS (0.420) but incurs HM = 131.63—1.25 px above RAWM v6 (Table 6).

5.3 Per-Horizon Analysis

RAWM v6 beats ZV at every horizon ($h = 1\text{--}5$) on both All and HM clips; the HM benefit grows monotonically from -7.1 px at $h = 1$ to -17.4 px at $h = 5$ ($\sim 10\%$ over ZV at the longest horizon, where proactive safety decisions are made). ZV error grows from 41 px at $h=1$ to 68 px at $h=5$ overall, and from 85 px to 174 px on HM clips; HM 95% CIs widen substantially at longer horizons (e.g., $[136, 233]$ px at $h=3$), reflecting the small HM test set ($N_{\text{HM}}=60$) and the high-variance nature of active driver behaviour.

5.4 Ablation Study

Table 5 shows all three components contribute. **No gate** ($+3.50$ px All): the gate inverts without regime supervision ($s_{\text{nhm}} > s_{\text{hm}}$: 0.388 vs. 0.333), which shows the BCE loss is essential. **No suppress** ($+4.89$ px All, largest): without ZV anchoring, the static branch drifts to arbitrary poses, degrading the $\approx 90\%$ non-HM clips. **No HM reweight** ($+4.62$ px All): class imbalance causes under-training of the rollout branch when all clips are weighted equally. As a further control, DRIVER-WM + HM $\times 5$ reweighting (no gate) achieves All-MPJPE = 75.79 px and VDS = 0.835, still failing the VDS test, confirming that loss reweighting alone cannot replace structural routing.

Table 6: Shrinkage baseline controls (test set, 136 joints, $N = 609$, $N_{\text{HM}} = 60$). [†]All-MPJPE not reported for oracle (GT routing label unavailable at inference). No global blend achieves both VDS < 0.5 and HM matching RAWM v6 simultaneously.

Config	All-MPJPE	HM-MPJPE	VDS
RAWM v6 (full, ours)	51.05	130.38	0.332
Global blend $\lambda=0.25$ (val-tuned)	51.23	130.31	0.502
Global blend $\lambda=0.20$	50.89	131.63	0.420
Oracle gate (GT label) [†]	—	134.89	0.028
Soft sigmoid (0 learned params)	50.83	130.84	0.429

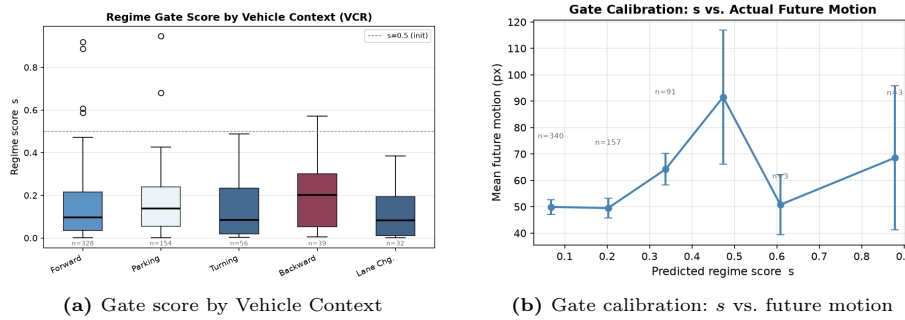


Fig. 2: Regime gate analysis (test set, $N = 609$). **Left:** gate assigns highest scores to *Backward Moving* ($s = 0.197$) and lowest to *Forward Moving* ($s = 0.136$) without label supervision. **Right:** gate score increases with future motion intensity (uncertainty indicator).

Shrinkage baseline controls. The oracle gate (GT routing labels) achieves HM=134.89 px—4.5 px worse than RAWM v6—confirming the HM gain derives from HM-reweighted training, not per-clip routing (Table 6). Among global blends, $\lambda=0.25$ matches HM (130.31) but fails VDS (0.502); $\lambda=0.20$ passes VDS (0.420) but trails HM (131.63); a soft sigmoid likewise (130.84 px); the gate Pareto-dominates both axes: VDS 0.332 (CI [0.29, 0.37]) vs. best blend 0.420, HM 130.38 vs. 131.63. Unlike a fixed blend requiring per-architecture λ search, the gate recipe generalises to three architectures and two datasets without retuning (Table 3).

5.5 Gate Interpretability

Fig. 2 shows the gate recovers motion semantics without explicit activity-label supervision: obs-window mean displacement classifies HM with AUC = 0.817 on AIDE (Section 3.3); the trained gate achieves AUC = 0.865 on Drive&Act (25 3D joints vs. 136 2D), showing kinematic routing generalises across coordinate systems. Vehicle-context scores reproduce the HM rate ordering from Section 3.1

Table 7: Regime gate applied to Simple MLP (pose-only, 1M params). VDS drops $0.801 \rightarrow 0.428$ without any VLM, camera, or architecture change; the gate works across a $193\times$ parameter range (Simple MLP 1M vs. RAWM 193M).

Model	Params	VLM	All-MPJPE	HM-MPJPE	VDS ($\downarrow$)
Simple MLP (no gate)	1M	$\times$	68.06	153.63	0.801
Simple MLP + gate (ours, AIDE)	1M	$\times$	53.64	155.28	0.428
Simple MLP + gate (D&A [‡])	1M	$\times$	0.041 m	0.148 m	0.458
DRIVER-WM	193M	$\checkmark$	71.47	138.03	0.883
RAWM v1 (ours)	193M	$\checkmark$	53.04	130.92	0.593
RAWM v6 (ours)	193M	$\checkmark$	51.05	130.38	0.332

[‡]D&A: metres (3D OpenPose, ZV: 0.041 m/0.151 m); gate: $3 \rightarrow 64 \rightarrow 32 \rightarrow 1$ (2.4K). Other rows: pixels (2D HALPE-136). Units differ; RAWM v6 VDS CI [0.29, 0.37].

without behaviour labels: *Backward Moving* ($s=0.197$, 21.1% HM) scores highest, *Forward Moving* ($s=0.136$, 8.0% HM) lowest.

5.6 Regime Gate is Model-Agnostic

We ask: does the regime gate transfer to a model it was not designed for, without VLM or camera features? We attach the gate ($<5K$ params) to the Simple MLP *unchanged* (Table 7): add only $\mathcal{L}_{\text{reg}}$ and the ZV blend $\hat{\mathbf{S}} = (1-s)\mathbf{S}_{\text{last}} + s\hat{\mathbf{S}}_{\text{MLP}}$.

All gate-augmented models pass VDS (Table 3). HM regression (+1.65 px) is expected since Simple MLP underperforms ZV on HM; RAWM’s rollout improves HM-MPJPE by 7.65 px. Body-joint-only VDS of RAWM v6 is 0.274 (vs. DRIVER-WM: 0.764), confirming the regime gain is not a face/hand keypoint artefact.

6 Limitations and Future Work

Gate pre-training (5-epoch frozen backbone, $\tau: 1.0 \rightarrow 0.05$) drives $s_{\text{nhm}} < 0.10$ stably; further improvement in s_{hm} (currently 0.15–0.28) via data augmentation would strengthen HM prediction. Hard-threshold routing ($< \tau_{90} \Rightarrow \text{ZV}$) passes VDS (0.028) but degrades HM-MPJPE to 181.67 px (+51 px vs. learned gate), confirming soft blending is necessary.

7 Conclusion

We identified the *motion regime imbalance problem*— $\approx 90\%$ low-motion clip dominance causes ERM to over-predict body motion (34% worse than ZV)—and proposed a **regime gate** ($<5K$ params) as the structural fix. The gate passes VDS across all tested architectures and datasets (Table 3); RAWM v6 beats ZV on All-MPJPE (-28.6% vs. DRIVER-WM), yielding a label-free uncertainty signal for proactive in-cabin safety.

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